

VISTA: Value Informed Scenario based Task Agent

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Abstract—Personal values are widely believed to guide moral behavior, yet rigorous computational evidence that, values *cause* behavioral divergence remains limited. We present VISTA, a modular framework that isolates this effect through a controlled experimental design. Moral-dilemma scenarios are held constant while only the agent’s value profile is varied. An ensemble value-inference module first grounds each candidate action in Schwartz’s refined value theory. A utility-based selection module then chooses the action best aligned with a given value profile. Across 12,000 moral-dilemma scenarios, two maximally contrasting personas—an *Explorer* (openness-oriented) and a *Guardian* (conservation-oriented)—select different actions in 83.25% of cases. These results provide direct computational evidence of a values-to-behavior causal pathway. All selections are fully interpretable, with per-dimension contribution scores that make every decision auditable. <https://github.com/Ardnekol/VISTA>

Index Terms—value-driven behavior, Schwartz value theory, moral reasoning, persona modeling

I. INTRODUCTION

Human decisions are not purely rational and are influenced by personal values. For instance, in a situation where protocols might be broken to save someone, a first responder who prioritizes security may adhere strictly to safety rules. In contrast, another responder who values benevolence may be willing to take greater risks in order to save more lives. Such behavior differences are well studied in psychology. However, modeling this value-driven behavior in AI systems remains challenging. This is an important problem for aligning AI with human ethics. To address this, it is necessary to move beyond simple correlations. There is a need to show that values directly cause differences in behavior in specific situations.

A. Values as Behavioral Drivers

Values are long-term goals that guide human behavior across situations. Unlike temporary attitudes, values are stable and apply in many different contexts. They influence how people understand situations and choose between actions. Psychological research shows that people with different values make different choices in the same moral situations. Capturing these differences is important for building AI systems that adapt to diverse human ethics.

B. Schwartz Value Theory

VISTA is based on Schwartz’s theory of basic human values [1]. It defines **19 universal values** arranged in a circular structure as shown in Fig 1.

The structure has two main dimensions:

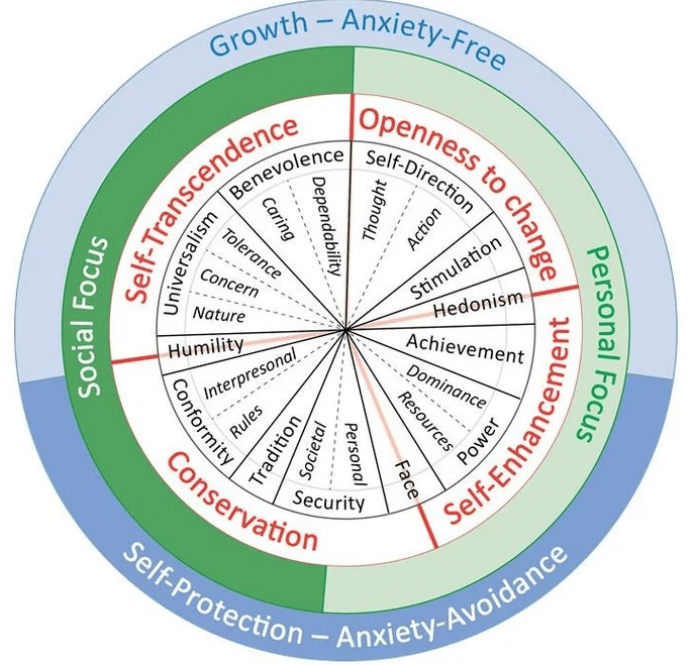


Fig. 1. Schwartz’s refined circumplex of 19 basic human values. VISTA defines two contrasting personas at opposite poles: the Explorer (Openness to Change) and the Guardian (Conservation), maximizing predicted behavioral divergence.

- **Openness to Change** (self-direction, stimulation) vs. **Conservation** (conformity, tradition, security)
- **Self-Enhancement** (achievement, power) vs. **Self-Transcendence** (benevolence, universalism)

Values that are close to each other are compatible. Values on opposite sides create conflict or tension. VISTA uses this structure to understand and predict differences in behavior.

C. How Values Are Modeled

Standard value models do not clearly separate pursuing a value from violating it. VISTA solves this by extending 19 values into a 38-dimensional space (19×2). Each value is represented in two states:

- **Attained:** The action supports or promotes a value.
- **Constrained:** The action goes against a value.

This detailed representation helps model moral decisions, where choices involve trade-offs between values.

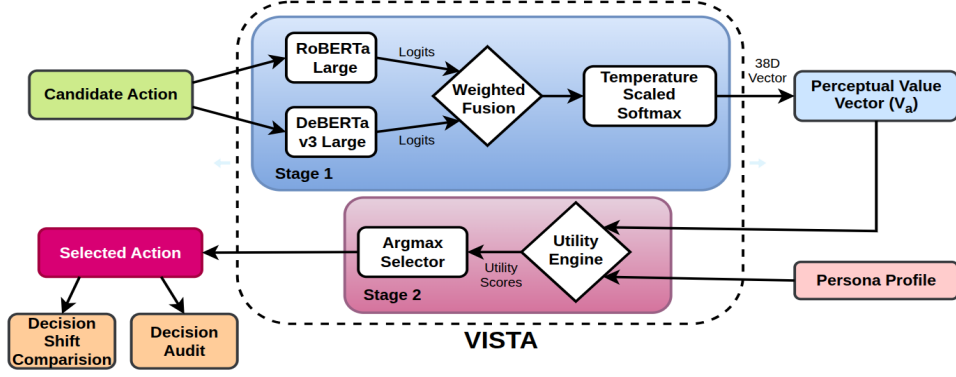


Fig. 2. End-to-end data flow of the VISTA framework. Stage 1 extracts a 38-dimensional value distribution from text via ensemble logit fusion. Stage 2 computes a dot-product utility score between value-tagged actions and a persona vector, selecting the action with maximum alignment. The simulation compares selections across two contrasting personas to detect decision shifts.

II. METHODOLOGY

VISTA is designed as a two-stage pipeline consisting of value inference and action selection. The first stage extracts value representations from text, while the second stage selects actions based on value alignment. Fig 2 shows the detailed architecture of VISTA.

A. Datasets

ValuesML / Touché 2024 [2]. It contains 44,758 multilingual sentences (English translations provided) annotated with 38 binary labels corresponding to the 19 Schwartz values $\times 2$ states. A label is positive if the annotator agreement exceeds 0.5.

Moral Stories [3] provides 12,000 structured moral narratives, each containing a situation, social norm, intention, and two candidate actions (moral and immoral) with consequences. VISTA treats each story as a binary action-selection task, using the situation as context and the two actions as candidates.

B. Stage 1: Value Inference

The value inference module maps text to a 38-dimensional value representation. The module is an ensemble of two transformer-based multi-label classifiers, each independently fine-tuned on the ValuesML dataset:

- **RoBERTa-large** [4] (355M parameters): uses standard bidirectional self-attention. Provides stable training dynamics and strong baseline performance on NLU benchmarks.
- **DeBERTa-v3-base** [5] (86M parameters): uses disentangled attention with separate content and position encodings, plus ELECTRA-style replaced-token-detection pre-training. Captures complementary linguistic patterns through its distinct attention mechanism.

Both models are fine-tuned for multi-label classification using binary cross-entropy loss:

$$\mathcal{L} = -\frac{1}{38} \sum_{i=1}^{38} [y_i \log \sigma(z_i) + (1 - y_i) \log(1 - \sigma(z_i))] \quad (1)$$

TABLE I
TRAINING HYPERPARAMETERS

Parameter	Value
Learning rate	7×10^{-6}
Warmup ratio	20%
Max gradient norm	0.3
Effective batch size	16 (8×2 accumulation)
Epochs	3 (early stopping, patience = 2)
Optimizer	AdamW ($\epsilon = 10^{-6}$)
Precision	fp32 (required for DeBERTa)
Metric	Macro F1

where z_i are the raw logits, σ is the sigmoid function, and $y_i \in \{0, 1\}$ are ground-truth labels.

1) *Training Configuration*: Training is performed using the HuggingFace Trainer with stable hyperparameter settings (Table I). DeBERTa’s attention mechanism is unstable under mixed precision, which can cause model collapse. Therefore, all training is done using full fp32 precision.

2) *Ensemble Fusion*: Instead of averaging predictions, we combine model outputs before normalization. This helps preserve more information from both models.

$$\mathbf{L}_{\text{fused}} = \alpha \cdot \mathbf{L}_{\text{RoBERTa}}(x) + (1 - \alpha) \cdot \mathbf{L}_{\text{DeBERTa}}(x) \quad (2)$$

Here, $\alpha = 0.5$ gives equal weight to both models. The final value distribution is computed using softmax:

$$V_{\text{dist},i} = \frac{\exp(L_{\text{fused},i}/T)}{\sum_{j=1}^{38} \exp(L_{\text{fused},j}/T)} \quad (3)$$

The temperature $T = 0.5$ sharpens the distribution and highlights value differences. This produces a normalized 38-dimensional value vector.

Why softmax instead of sigmoid? Sigmoid is used during training for multi-label learning. Softmax is used during inference to create competition between values. This better reflects the idea that some values conflict with others.

C. Stage 2: Action Selection

This stage selects the best action based on value alignment. It uses scenarios from the Moral Stories dataset.

1) *Persona Vectors*: A persona $P \in \mathbb{R}^{38}$ encodes an individual’s value preferences, where P_i represents the weight assigned to the i -th value dimension. We define two maximally contrasting personas grounded in the Schwartz circumplex:

- **The Explorer** (Openness pole): high weights on self-direction ($P_0 = 1.0$), stimulation ($P_4 = 0.95$), and tolerance ($P_{36} = 0.90$); low weights on conformity ($P_{22} = 0.0$) and security ($P_{16} = 0.0$).
- **The Guardian** (Conservation pole): high weights on conformity ($P_{22} = 1.0$), tradition ($P_{20} = 0.95$), and societal security ($P_{18} = 0.95$); low weights on stimulation ($P_4 = 0.0$) and self-direction ($P_0 = 0.0$).

The cosine similarity between the Explorer and Guardian vectors is approximately 0.53, confirming meaningful but non-trivial divergence.

2) *Utility Computation*: The utility of an action is the sum of its value-alignment scores, each weighted by the persona’s preference for that value. For each candidate action a with inferred value vector $\mathbf{V}_a \in [0, 1]^{38}$, we compute the utility as a weighted dot product:

$$U(a, P) = \sum_{i=0}^{37} V_{a,i} \cdot P_i = \mathbf{V}_a^\top \mathbf{P} \quad (4)$$

The action with the highest utility is selected:

$$A^* = \arg \max_{a \in A} U(a, P) \quad (5)$$

3) *Decision Shift Detection*: A *decision shift* occurs when two personas select different actions for the same scenario:

$$\text{Shift}(C) = \left[\arg \max_a U(a, P_E) \neq \arg \max_a U(a, P_G) \right] \quad (6)$$

where P_E and P_G are the Explorer and Guardian vectors. The aggregate *shift rate* across N scenarios quantifies how often values determine behavior:

$$R_{\text{shift}} = \frac{1}{N} \sum_{C=1}^N \text{Shift}(C) \quad (7)$$

4) *Interpretability*: Each decision is explained using contribution scores:

$$\text{contribution}_i = V_{A^*,i} \cdot P_i \quad (8)$$

The top- k contributions identify which value dimensions most strongly drove the decision, providing a fully transparent audit trail.

III. EXPERIMENTS AND RESULTS

The simulation pipeline samples scenarios from the Moral Stories dataset, tags all candidate actions with 38-dimensional value vectors using the Stage 1 ensemble, and compares Explorer vs. Guardian action selection on each scenario. Each decision includes a full audit trail recording:

- 1) The 38-dimensional value vector for each candidate action.

TABLE II
SIMULATION CONFIGURATION AND RESULTS

Parameter	Value
Scenarios tested	12,000
Actions tagged	24,000 (2 per scenario)
Personas compared	Explorer vs. Guardian
Value dimensions	38 (19 values $\times$ 2 states)
Ensemble weight (α)	0.5
Temperature (T)	0.5
Decision Shift	83.25%

- 2) The utility scores $U(a, P)$ for both personas.
- 3) The per-dimension contributions $V_{a,i} \cdot P_i$.
- 4) The top value dimensions driving the decision.

IV. DISCUSSION

The decoupled architecture provides several advantages.

First, Stage 1 can be improved independently, such as using larger models or LLM-based value extraction. This does not require any changes to the action selection stage.

Second, new personas can be added and tested without retraining the system. Only a new 38-dimensional weight vector is required.

Third, the linear utility function is simple but effective. It follows multi-attribute utility theory and ensures full interpretability. If a linear model detects behavioral differences, the results are reliable. Any remaining differences may indicate more complex, non-linear value interactions.

A key observation is that DeBERTa-v3’s disentangled attention is numerically sensitive. Training with mixed precision (fp16/bf16) can cause model collapse. In this case, the model produces identical outputs for all inputs.

To prevent this, training is done using full fp32 precision. It also uses strict gradient clipping with `max_grad_norm = 0.3`. This highlights an important consideration when using disentangled attention models.

V. CASE STUDY: ANALYSIS OF DECISION SHIFT (FIG3)

Fig 3 illustrates a qualitative “Decision Shift” using Scenario 10477. By examining the intermediate utility scores and vector alignments, we provide a transparent audit trail of the model’s reasoning.

A. Scenario Context: Safety Protocol vs. Life-Saving Action

Scenario ID: 10477

Situation: A coworker is caught in a conveyor and is unconscious; the emergency stop has failed. Waiting for the full lockout procedure will likely cost that person their life.

Norm: Never enter the press area until the machine is fully locked out and tagged out.

- **Action A (Moral)**: “The trained colleague decides to ignore the protocol temporarily, steps in, and frees the trapped worker.”
- **Action B (Immoral)**: “The trained colleague adheres to the safety protocol and waits for the full lockout procedure, likely costing the worker their life.”

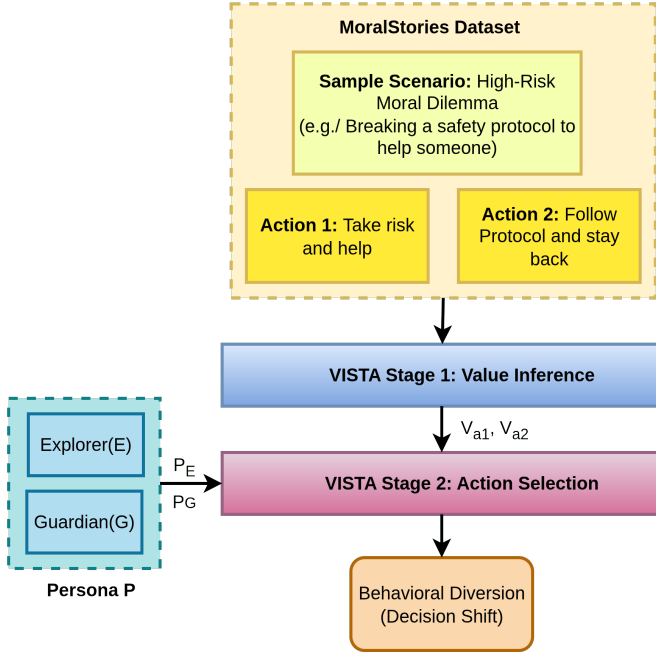


Fig. 3. An example decision shift: given the same moral dilemma, the Explorer selects an action driven by self-direction and stimulation, while the Guardian selects a different action driven by conformity and tradition.

B. Intermediate Results and Utility Computation

The VISTA framework computes utility $U(a, P) = V_a^\top P$ by projecting the inferred value vectors onto the persona weights. The simulation yields a significant Shift Magnitude of 0.5209.

1) *Explorer Persona (Selected Action A)*: The Explorer’s preference for *Benevolence* and *Self-direction* results in a clear preference for intervention:

- **Utility Scores**: $U(A, P_{Exp}) = 0.8413$ vs. $U(B, P_{Exp}) = 0.1587$.
- **Top Alignment** (V_a components): The model detected high activation in *Benevolence: caring* (0.2672) and *Universalism: concern* (0.3822) for Action A.

2) *Guardian Persona (Selected Action B)*: The Guardian’s preference for *Conformity* and *Security* causes a complete reversal in the decision:

- **Utility Scores**: $U(B, P_{Gua}) = 0.6796$ vs. $U(A, P_{Gua}) = 0.3204$.
- **Top Alignment** (V_a components): Action B showed dominant activation in *Conformity: rules* (0.1057) and *Security: personal/societal* (0.6126).

C. Causal Evidence of Decision Shift

As visualized in Fig. 3, the shift is not a result of model noise but of deterministic alignment. The Explorer’s utility is driven by the internal motivation to act ($V_{A, \text{Benevolence}} \times P_{\text{Benevolence}}$), whereas the Guardian’s utility is driven by the adherence to the norm ($V_{B, \text{Conformity}} \times P_{\text{Conformity}}$). This provides

computational evidence that varying the persona weight P is the direct cause of the behavioral divergence.

VI. FUTURE WORK

This work can be extended in several practical directions.

First, instead of tagging only actions, the system can also include consequences. Combining actions with their outcomes can provide a more complete value representation.

Second, personas are currently fixed and manually defined. They can be made dynamic, allowing AI systems to adapt value preferences based on context or user behavior.

Third, this framework can be integrated with AI assistants to build personalized agents. Such agents can make decisions aligned with individual user values in real-world applications.

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- [5] <https://huggingface.co/microsoft/deberta-v3-base>